

Bonus questions: Sample Spaces, σ -Algebras and Probability

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Issued: 21 August 2026

• Due: Self-study

B1. Buffon's needle. A plane contains parallel lines at distance 2ℓ from one another. A needle of length ℓ is thrown onto the plane. Let r be the distance from the centre of the needle to the nearest line, and let θ be the angle between the needle and the parallel lines. We take

$$0 \leq r \leq \ell, \quad 0 \leq \theta \leq \pi,$$

and assume that (r, θ) is uniformly distributed over this rectangle.

(a) Show that the needle crosses a line exactly when

$$r \leq \frac{\ell}{2} \sin \theta.$$

(b) Draw this event in the (θ, r) -plane and find its area.

(c) Divide by the area of the whole sample space and show that

$$P(\text{needle crosses a line}) = \frac{1}{\pi}.$$

This gives an experiment that can be used to estimate π .

B2. Bertrand's paradox. A chord is drawn “at random” in a circle of radius 1. Find the probability that the chord is longer than the side of an equilateral triangle inscribed in the circle, using each of the following meanings of “at random”.

(a) Choose the midpoint of the chord uniformly from the disk.

(b) Fix one endpoint and choose the other endpoint uniformly from the circumference.

(c) Choose a radius, and choose the distance of the chord's midpoint from the centre uniformly from $[0, 1]$; draw the chord perpendicular to the radius.

You should get three different answers. Explain why this is not a contradiction. What changed from one experiment to another?

B3. Derangements: letters and envelopes. A secretary has n letters addressed to n different people, and n envelopes carrying the corresponding addresses. The letters are placed into the envelopes at random, one letter in each envelope. Assume that all $n!$ arrangements are equally likely.

Let A_i be the event that letter i is placed in the correct envelope.

(a) Find

$$P(A_i), \quad P(A_i \cap A_j), \quad P(A_{i_1} \cap \cdots \cap A_{i_k}).$$

(b) Use the principle of inclusion–exclusion to show that the probability that at least one letter is placed in the correct envelope is

$$1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1} \frac{1}{n!}.$$

(c) Hence show that the probability that no letter is placed in the correct envelope is

$$\sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Such an arrangement is called a *derangement*.

(d) What happens to this probability as $n \rightarrow \infty$?

B4. The Monty Hall problem. A game show has three closed doors. Behind one door is a car, and behind the other two are goats.

The contestant first chooses one door, but the door is not opened. The host knows where the car is. He then opens one of the other two doors, always choosing a door with a goat behind it. If both doors available to him contain goats, he chooses one of them at random.

The contestant is now allowed either to *stay* with the original choice or to *switch* to the other unopened door.

- (a) Before the host opens a door, what is the probability that the contestant's first choice contains the car?
- (b) Suppose the contestant always stays with the first choice. What is the probability of winning the car?
- (c) Suppose the contestant always switches after the host opens a door. What is the probability of winning the car?
- (d) Which strategy is better?
- (e) Explain why the host opening a door is not the same as simply removing one of the two unchosen doors at random.

Optional challenge. Write down a sample space that includes the position of the car, the contestant's first choice, and the door opened by the host. Assign probabilities to its outcomes according to the rules above, and verify your answers directly.

A generalized Monty Hall problem. Suppose instead of 3 doors, we have 100 doors, concealing one car and ninety-nine goats. Again, the contestant chooses any door, and this time Monty opens ninety-eight other doors to reveal goats behind them all. Now, should you switch your choice to the last remaining door?

B5. Bonus: A recipe for a Vitali set.

We begin by recalling some ideas about quotient groups. Let G be an abelian group and let H be a subgroup of G . The quotient

$$G/H = \{g + H : g \in G\}$$

is the collection of all "cosets" of H in G . Two elements $g, g' \in G$ give the same coset precisely when

$$g + H = g' + H \iff g - g' \in H.$$

The set G/H is itself an abelian group under the operation

$$(g + H) + (g' + H) := (g + g') + H.$$

- (a) Check that the above operation is well-defined. That is, if

$$g + H = x + H \quad \text{and} \quad g' + H = x' + H,$$

show that

$$(g + g') + H = (x + x') + H.$$

For example, take

$$G = \mathbb{Z}, \quad H = 5\mathbb{Z}.$$

Then the quotient

$$\mathbb{Z}/5\mathbb{Z}$$

consists of the five cosets

$$5\mathbb{Z}, \quad 1 + 5\mathbb{Z}, \quad 2 + 5\mathbb{Z}, \quad 3 + 5\mathbb{Z}, \quad 4 + 5\mathbb{Z}.$$

Each coset consists of all integers which leave a fixed remainder when divided by 5.

A *complete system of representatives* for G/H is obtained by choosing exactly one element from each coset. For example,

$$\{0, 1, 2, 3, 4\}, \quad \{5, 1, 2, 3, 4\}, \quad \{0, 6, -3, 3, 9\}$$

are all complete systems of representatives for $\mathbb{Z}/5\mathbb{Z}$.

- (b) Check that each of the three sets above is indeed a complete system of representatives for $\mathbb{Z}/5\mathbb{Z}$.

- (c) Let S be any complete system of representatives for G/H . Show that every $g \in G$ can be written uniquely in the form

$$g = s + h, \quad s \in S, \quad h \in H.$$

- (d) Deduce that G is partitioned by the translates of S by elements of H :

$$G = \bigsqcup_{h \in H} (S + h),$$

where

$$S + h = \{s + h : s \in S\}.$$

We now apply this idea to the interval

$$[0, 1).$$

We equip $[0, 1)$ with addition modulo 1. Thus, for $x, y \in [0, 1)$, define

$$x \oplus y := \{x + y\},$$

where $\{t\}$ denotes the fractional part of t . For example,

$$0.7 \oplus 0.6 = 0.3.$$

With this operation, $[0, 1)$ is an abelian group; call it G .

- (e) Check that $[0, 1)$, with the operation $\oplus$, is an abelian group. What are its identity element and the inverse of an element $x \in [0, 1)$?

Now let

$$H = \mathbb{Q} \cap [0, 1),$$

with the same operation of addition modulo 1. Consider the quotient group

$$G/H.$$

Choose exactly one element from every coset of H (**alert:** we can make such a choice from every coset at once by the Axiom of Choice), and let $S \subseteq [0, 1)$ be the set of chosen elements. Thus S is a complete system of representatives for G/H . Such a set is called a *Vitali set*.

- (f) Show that the sets

$$S \oplus h := \{s \oplus h : s \in S\}, \quad h \in H,$$

are pairwise disjoint.

- (g) Show that

$$[0, 1) = \bigsqcup_{h \in H} (S \oplus h).$$

We now ask what would happen if we tried to define the uniform probability on

$$([0, 1), \mathcal{P}([0, 1))),$$

so that *every* subset of $[0, 1)$ is an event. Informally, $P(A)$ should give the “length” of a set $A \subseteq [0, 1)$. It is reasonable to demand that translation of a set modulo 1 should not change its probability, i.e., for any event A we should have $P(A) = P(x \oplus A)$.

Suppose that

$$P(S) = p_0.$$

Since H is countably infinite, write

$$H = \{h_1, h_2, h_3, \dots\}.$$

By the previous parts,

$$[0, 1) = \bigsqcup_{i=1}^{\infty} (S \oplus h_i).$$

(h) Explain why

$$P(S \oplus h_i) = p_0$$

for every i .

(i) Use countable additivity to show that

$$1 = P([0, 1)) = \sum_{i=1}^{\infty} P(S \oplus h_i) = \sum_{i=1}^{\infty} p_0.$$

(j) Explain why the series

$$\sum_{i=1}^{\infty} p_0$$

can converge only when $p_0 = 0$, and why $p_0 = 0$ also gives a contradiction.

(k) Conclude that $P(S)$ cannot be defined. Hence explain why we cannot take the whole power set

$$\mathcal{P}([0, 1))$$

to be the collection of events for the uniform probability on $[0, 1)$, and more generally why we do not take every subset of an interval in $\mathbb{R}$ to be an event.